



SOUTHEAST UNIVERSITY
Department of Computer Science and Engineering (CSE)
Advanced Java Lab (CSE352)
Summer – 2026 | Section 3,4
Assignment-1

AOOP Assignment: Java Collections & I/O with REST API

Marks: 15

Course: Advanced Object-Oriented Programming (AOOP) **Tools:**

Java (REST API) and Postman

Assignment Objective:

Develop a **Java-based RESTful Student Management System** that demonstrates the use of the **Collection Framework, Comparable & Comparator, Iterator, Set and Stack**, and **Java I/O with Serialization**. All API operations must be tested using **Postman**.

Requirements:

1. Core Implementation:

- ✚ Create a Student class with: id, name, department, cgpa.
- ✚ Store students in an **ArrayList**.
- ✚ Implement REST endpoints to:
 - Add a student.
 - View all students (using **Iterator**).
 - Sort by **ID** (Comparable).
 - Sort by **CGPA (descending)** (Comparator).
- ✚ Test all endpoints using Postman.

2. Additional Collections:

- ✚ Use a **HashSet** to return **unique departments**.
- ✚ Use a **Stack** to manage the **last three added students** (push, pop, peek).

3. File I/O & Serialization (5 Marks):

- ✚ Implement **serialization** to save student objects to a file.
- ✚ Implement **deserialization** to restore data from the file.
- ✚ Briefly explain **Byte Streams vs. Character Streams** with examples in the report.

Submission:

- ✚ Source code.
- ✚ Postman screenshots.
- ✚ Short report (including stream comparison).